

Exercises

Exercises on *foldr*

Exercise 1:

Using the higher-order function *foldr*, define a function *sumsq* which takes an integer n as its argument and returns the sum of the squares of the first n integers. That is to say, $sumsq\ n = 1^2 + 2^2 + 3^2 + \dots + n^2$.
(answer given to start you off)

Exercise 2:

Define *lengthr*, which returns the number of elements in a list, using *foldr*.

Exercise 3:

Define *minlist*, which returns the smallest integer in a non-empty list of integers, using *foldr1*. (*foldr1* is a Prelude function - look it up yourself or continue and come back to this)

Exercise 4:

Define *myreverse*, which reverses a list, using *foldr*.

Exercise 5:

Using *foldr*, define a function *remove* which takes two strings as its arguments and removes every letter from the second list that occurs in the first list. For example,

```
remove "first" "second" = "econd".
```

Hint: Use a helper function in your lambda

Exercise 6:

The function *remdups* removes adjacent duplicates from a list. For example,

```
remdups [1, 2, 2, 3, 3, 3, 1, 1] = [1, 2, 3, 1]
```

Define *remdups* using *foldr* (and using a helper method).